

RESEARCH ARTICLE

A survey of westward-propagating mixed Rossby–Gravity waves and quantification of their association with extratropical disturbances

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Abstract

In this study we have conducted a survey of westward-propagating mixed Rossby–Gravity (MRG) wave events in the upper troposphere and quantified their association with the intrusions of extratropical disturbances for the period 1979–2019. MRG wave events are identified by imposing the meridional structure of theoretical MRG waves onto the equatorial meridional winds at 200 hPa. In all, 2390 MRG wave events are identified and the majority (61%) of them occurred during the months of May–October, and 65% of the total MRG wave events occurred over the central–east Pacific and Atlantic Ocean domains. Not only the frequency of occurrence but also the amplitude, wavenumber and trapping scale of the MRG wave events are found to exhibit a clear seasonality. MRG wave events associated with intrusions of extratropical disturbances are identified as when the potential vorticity on the 350-K isentropic surface at 15° latitude exceeded 1 Potential Vorticity Unit (PVU) in the vicinity of the MRG wave events. We find that 37% of the MRG wave events are intrusion MRG wave events and a large majority (88%) of such events occurred over the central–east Pacific and Atlantic Ocean domains. It is also noteworthy that nearly 70% of such intrusions occurred in the winter hemisphere where the westerly wind ducts are well developed. Over the central–east Pacific during northern hemispheric (NH) winter, it is observed that the intrusion MRG wave events have a bigger amplitude and have a larger meridional extent compared to non-intrusion MRG wave events. They also exhibit a similar spatial scale as the extratropical disturbances implying that resonant interactions may be a primary mechanism for the genesis of MRG wave events. During NH summer, on the other hand, MRG wave events are primarily triggered by convective processes and the extratropical disturbances may be instrumental in amplifying their amplitude.

KEYWORDS

equatorial waves, mixed Rossby–Gravity waves, synoptic-scale variability, tropical–extratropical interactions

1 | INTRODUCTION

Mixed Rossby–Gravity (MRG) waves are synoptic-scale westward-propagating disturbances observed in dynamic and convective fields over the equatorial region (Hendon & Liebmann, 1991; Kiladis et al., 2016; Suhas et al., 2020; Takayabu, 1994a; Takayabu & Nitta, 1993; Takayabu et al., 2016; Wheeler & Kiladis, 1999). Pronounced cross-equatorial meridional winds and vortices distributed symmetrically about the equator are the prominent features of MRG waves. MRG waves are more frequently observed over the Pacific domain during boreal summer and fall (Hendon & Liebmann, 1991; Kiladis et al., 2016; Magaña & Yanai, 1995; Suhas et al., 2020; Yasunaga, 2011). Several studies have shown that the MRG waves are sensitive to the dynamic and convective background fields in which they are embedded (Wang & Xie, 1996; Zhang & Webster, 1989, 1992). MRG waves are found to be more frequent during the La Niña phase of the El Niño Southern Oscillation (ENSO) and they are confined to the upper troposphere when there is westerly vertical shear or to the lower troposphere when there is easterly vertical shear (Feng et al., 2016; Kiladis et al., 2016; Takayabu & Nitta, 1993; Wang & Xie, 1996).

MRG waves play a pivotal role in dictating the variability and predictability of tropical weather. For example, MRG waves originating over the central or east Pacific often transition to tropical depressions and cyclones over the west Pacific (WP) (Aiyer & Molinari, 2003; Feng et al., 2016; Takayabu & Nitta, 1993; Zhou & Wang, 2007). Recent studies indicate that MRG dynamics play a major role in initiating large-scale organised convection over the Indian Ocean (IO) (Takasuka et al., 2019; Takasuka & Satoh, 2020, 2021). Contribution of easterly momentum via vertically propagating MRG waves is vital for the quasi-biennial oscillation (QBO), a dominant mode of stratospheric circulation (Holton & Lindzen, 1972; Yanai et al., 1968). MRG waves were identified to be one of the factors that caused the disruption in QBO activity during 2015–2016 period (Lin et al., 2019). Recent theoretical and observational studies point out that dry dynamics is as important as moist dynamics in explaining the initiation, structure and propagation of the Madden–Julian Oscillations (MJO) and MRG wave is a core part of such theories (Pritchard & Yang, 2016; Takasuka et al., 2019; Takasuka & Satoh, 2020; Yang & Ingersoll, 2011). Since MRG waves bear the character of both Rossby and inertia–gravity waves, understanding the dynamics of MRG waves is crucial for improving the method of initialisation and thereby the skill in predicting tropical weather.

Since the advent of satellite observations, many studies have been conducted to understand the structure, propagation and excitation mechanisms of MRG waves

(Hendon & Liebmann, 1991; Kiladis et al., 2016; Takayabu & Nitta, 1993; Yanai & Lu, 1983; Zangvil & Yanai, 1981; Zhang & Webster, 1992). Tropical convection, lateral forcing of extratropical disturbances, underlying sea surface temperature (SST) configuration, hydrodynamical instability and wave–wave interactions are suggested as possible mechanisms that can excite MRG waves (Hendon & Liebmann, 1991; Kiladis et al., 2009; Magaña & Yanai, 1995; Randel, 1992; Ripa, 1983). Many of the studies that tried to address the properties of MRG waves and their interactions with other modes of variability often focused on the convectively coupled MRG waves (Kiladis et al., 2016; Suhas et al., 2020; Wheeler et al., 2000; Wheeler & Kiladis, 1999). However, not all MRG wave activity may have a direct association with convection. For example, over the eastern Pacific and Atlantic domains, even while the synoptic-scale perturbation kinetic energy is high, the large-scale convection–circulation coupling strength is weak and hence a large fraction of MRG waves over these regions may not have a direct association with convection (Chen et al., 2016; Yang & Hoskins, 2016). Theoretical and observational studies recognise extratropical wave intrusions as one of the main sources of synoptic-scale variability over such regions where the probability of the occurrence of organised convection is less (Charney, 1963; Tomas & Webster, 1994; Webster & Holton, 1982). Linear wave theory identifies the presence of the upper-tropospheric westerly wind ducts as an important condition that would permit the intrusions of extratropical disturbances (Webster & Holton, 1982). Many observational studies show that extratropical intrusions through the upper-tropospheric westerly ducts can trigger tropical convection and equatorial waves (Kiladis, 1998; Yang & Hoskins, 2016). Wave energy accumulation and dispersion of energy from other locations can also induce synoptic-scale variability over regions where organised convective activity is less active (Hoskins & Yang, 2016; Webster & Chang, 1988; Yang & Hoskins, 2016). Hence a convection-based (e.g., using outgoing longwave radiation [OLR], precipitation, brightness temperature, etc.) identification of MRG wave activity would be an underestimate, as there might be several MRG wave events that might not have a strong signature in convective fields. Therefore, in order to have a comprehensive understanding of the MRG wave activity and their properties, it is important to use a method that is equally sensitive to all categories of MRG waves.

In this study we plan to conduct a climatological survey of MRG waves and characterise them over the whole tropics and separately over different geographical domains. Several past studies that brought out the characteristics of the MRG waves adopted a case study-based approach, which were mostly conducted over localised regions using

data of limited sample size (Hendon & Liebmann, 1991; Liebmann & Hendon, 1990; Takayabu, 1994b; Takayabu & Nitta, 1993). Another set of studies extracted the climatological properties of MRG waves over a wider domain using a combination of wavenumber–frequency filtering and linear regression methods (Kiladis et al., 2009, 2016; Suhas et al., 2020; Wheeler et al., 2000). Since the observed MRG waves are localised in space and time, a Fourier-based wave isolation method might not be ideal for bringing out the local characteristics of the waves, as Fourier transform employs the assumption that the MRG waves are global in nature and are continuous. Alternately, wavelet and windowed wavenumber–frequency spectral analysis are used to capture the localised equatorial wave events (Dias & Kiladis, 2014; Kikuchi, 2014). However, these approaches are also not very effective in capturing the episodic MRG events. In order to better capture the spatial and temporal inhomogeneous nature of MRG wave activity, we use an event-based MRG wave identification method proposed by Au-Yeung and Tam (2018).

Since the main objective of the study is to characterise the MRG waves of all kinds, we chose the 200-hPa meridional wind as the primary variable to identify the MRG wave events. The upper-tropospheric winds capture both convectively coupled MRG waves over the Pacific and IO domains and dry MRG waves over the central and east Pacific and Atlantic domains (Kiladis et al., 2016; Magaña & Yanai, 1995; Randel, 1992; Wheeler & Kiladis, 1999; Yanai & Lu, 1983; Yang et al., 2003). The MRG signal in the lower atmosphere is mainly contaminated by the tropical depressions (TD), as it is very difficult to separate out these disturbances (Au-Yeung & Tam, 2018; Takasuka & Satoh, 2020; Takayabu & Nitta, 1993). However, the MRG event identification method based on upper-tropospheric winds is less likely to be contaminated by TD, as the influence of TD is largely confined to the lower to middle troposphere (Feng et al., 2016). Recent studies bring out observational evidence for the role of upper-tropospheric MRG waves in triggering lower-level MRG waves and their subsequent transition to TD and tropical cyclones (Au-Yeung & Tam, 2018; Zhou & Wang, 2007). The intrusion of extratropical disturbances over certain tropical regions is also considered as a triggering mechanism for MRG waves (Kiladis et al., 2016; Magaña & Yanai, 1995; Mak, 1969; Yang & Hoskins, 2016; Zhang & Webster, 1992). According to linear wave theory, extratropical disturbances are more likely to intrude over tropical regions where the background winds are westerly (Webster & Holton, 1982). There exists a direct link between the strength of the mean westerly winds and the intensity of energy carried by the extratropical disturbances. The observation of a stronger westerly wind duct in the upper troposphere over the central and east

Pacific and Atlantic domains makes 200-hPa meridional wind an ideal parameter for quantifying MRG events associated with extratropical intrusions. Although many previous studies brought out observational evidence for the role of extratropical forcing in triggering MRG waves (Kiladis et al., 2016; Magaña & Yanai, 1995; Randel, 1992; Yang & Hoskins, 2016; Zangvil & Yanai, 1981), the relative contribution of such a forcing has not been quantified.

It is expected that a detailed climatology of the MRG events and their characterisation over different geographical domains will provide an observational basis for a complete assessment of MRG excitation mechanisms. It will also serve as a testbed for evaluating the skill of climate models in simulating the MRG waves. Since the MRG waves are unique to the tropics and they play an active role in tropical-scale interactions, realistic simulation of MRG waves is crucial for ensuring the reliability of models in simulating the weather and climate variability. It has been noted that extratropical disturbances could modulate the tropical synoptic-scale variability (Charney, 1963; Mak, 1969). A recent study by Vishnupriya et al. (2022) showed that about 15% of the monsoon low-pressure systems are triggered by extratropical air intrusions over the Pacific. A set of recent studies revealed that equatorward-propagating extratropical Rossby waves from the Northern (NH) and Southern (SH) hemispheres and their association with MRG waves contribute nearly half of the wave forcing that caused disruption in the westerly phase of the QBO (Kang et al., 2020; Kang & Chun, 2021; Lin et al., 2019). These results demonstrate the close association between tropical and extratropical dynamics and highlight the importance of quantifying the contribution of extratropical disturbances in modulating the MRG waves. Since QBO is considered a source of predictability on the seasonal time-scales, a comprehensive understanding on the relative role of extratropical disturbances in modulating the MRG wave events is also crucial for improving the prediction skill of extratropical seasonal forecasts (Gray et al., 2018). We believe that analysis of MRG events using 40 years of reanalysis data will help address the long-standing issue of the degree of involvement of extratropical disturbances in determining the MRG waves in the tropics.

2 | DATA AND METHODOLOGY

European Centre for Medium-Range Weather Forecasts Re-Analysis (ERA-Interim) six-hourly $0.5^\circ \times 0.5^\circ$ horizontal-resolution zonal and meridional components of the winds at 200 hPa and daily $2.5^\circ \times 2.5^\circ$ horizontal-resolution potential vorticity (PV) on the 350-K isentropic surface from 1 January 1979 to 31 August 2019

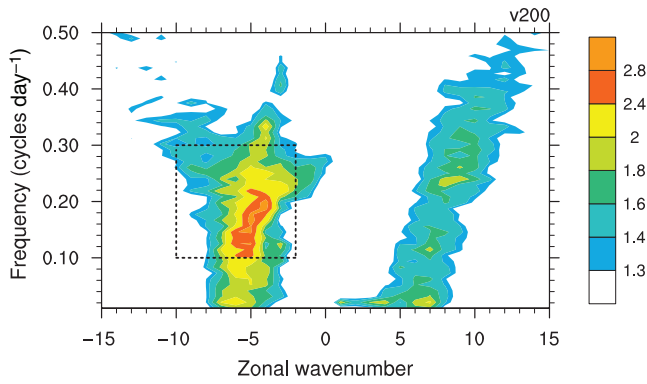


FIGURE 1 Symmetric/background zonal wavenumber–frequency power spectra of 200 hPa meridional wind (v_{200}) in the tropics (15°S – 15°N). The black dashed box isolates the wavenumbers and frequencies corresponding to the westward-propagating mixed Rossby–Gravity (MRG) wave activity. Shaded regions are statistically significant at the 99% confidence level. [Colour figure can be viewed at wileyonlinelibrary.com]

are the primary data used in this study (Dee et al., 2011). Figure 1 shows the symmetric wavenumber–frequency power spectra of 200-hPa meridional wind in the tropics (15°S – 15°N) normalised by the red background (Wheeler & Kiladis, 1999). MRG wave power is prominently distributed in westward-propagating wavenumbers -10 to -2 in the 3–10-day period. It is interesting to note the presence of significant spectral peaks in the eastward wavenumbers as well. Some part of the eastward spectral power might be a result of Doppler shifting of $n=2$ equatorial Rossby (ER) and MRG waves by the zonal flow (Yang et al., 2023; Yang & Hoskins, 2016) and some fraction of the eastward spectral power might be contributed by the intrusion of extratropical disturbances (Kiladis, 1998). Even though the spectral power associated with eastward-propagating MRG waves is less than that associated with westward-propagating MRG waves, it contributes to a large fraction of the eastward-propagating synoptic-scale variability in the western hemisphere (Yang & Hoskins, 2016). However a proper investigation of eastward-propagating MRG waves demands a separate study and in this study, we focus on westward-propagating MRG waves. In the following sections westward-propagating MRG waves are referred to simply as MRG waves.

MRG wave events at the 200-hPa level are identified by following the methodology of Au-Yeung and Tam (2018). The method is ideal for capturing the episodic behaviour of the MRG wave events. Westward-propagating strong equatorial meridional winds are the typical signature of the MRG waves and the MRG wave event identification methodology was designed by exploiting this property. The method estimates the instantaneous properties of the

MRG waves such as the amplitude, trapping scale, frequency, wavelength and phase speed during the entire duration of an event. The crux of the methodology is based on the imposition of the theoretical meridional structure of MRG waves onto the meridional wind anomalies (Matsuno, 1966). The meridional wind anomalies are estimated by filtering meridional winds in the westward wavenumbers 2–10 and 3–10-day periodicity. Since the meridional winds associated with MRG waves are symmetrically distributed about the equator, only the symmetric meridional wind anomalies are imposed onto the theoretical meridional structure. The MRG event identification methodology involves multiple steps. At first, the longitudes marking the centres of the MRG wave events are located by identifying the local maxima and minima of the meridional wind anomaly at the equator for each time instant. Only the longitudes where the meridional wind anomalies exceed a value equal to 20% of the maximum, meridional wind amplitude are recorded to avoid very weak MRG wave events. Secondly, the meridional structure of the MRG waves is estimated at all those longitude locations by fitting the meridional wind anomalies at the equator onto the theoretical meridional structure of MRG waves using the equation given below.

$$v_{\text{fitted}}(x, y, t) = v_{\text{data}}(x, y = 0^{\circ}, t) D_0(y), \quad (1)$$

$$D_0(y) = e^{-\left(\frac{y}{y_0}\right)^2}, \quad (2)$$

where v_{fitted} is the meridional wind of MRG waves and v_{data} is the meridional wind anomaly, x, y, t represent the longitude, latitude and time points, respectively, $D_0(y)$ is the $n=0$ parabolic cylindrical function which describes the meridional structure of the $n=0$ equatorial wave mode (Yang et al., 2003) and y_0 represents the trapping scale or idealised representation of meridional extent of the MRG wave. y_0 is a free parameter and MRG meridional winds are estimated for y_0 ranging from 0.5° to 20° .

The error associated with the estimation of MRG meridional winds is quantified as follows:

$$E(x, t) = \frac{\sqrt{\frac{1}{n} \sum_{y=0^{\circ}}^{y=20^{\circ}} (v_{\text{data}}(x, y, t) - v_{\text{fitted}}(x, y, t))^2}}{|v_{\text{data}}(x, y = 0^{\circ}, t)|} \quad (3)$$

where E represents the error as a function of longitude and time and n represents the number of latitude points between 0° and 20° .

Finally, the observed maximum (minimum) of meridional wind anomaly at a longitude and time step is considered as a part of an MRG event only if it is linked to the next time step in a way that the maximum (minimum) of meridional wind anomaly should be observed within 20°

west of the previous longitude point. A cluster of longitude and time points that satisfy the above criteria for a duration ranging from two to eight days with at least one point exhibiting an error value less than or equal to 5% is classified as an MRG wave event. To avoid double counting, we have identified the MRG wave events by considering only the positive phase of the meridional wind anomaly.

Even though the wave-fitting method employed in the study is derived from the wave projection methodology of Yang et al. (2003), there are significant differences between the two approaches. In the wave projection method parabolic cylindrical functions are used as the basis function and since the parabolic cylindrical functions are orthogonal, the projection of dynamical fields onto the theoretical structure of equatorial waves does not permit the mixing of different equatorial waves, whereas in the wave-fitting method, the imposition of meridional wind anomalies¹ onto the meridional structure of MRG waves, while capturing the equatorially trapped and equatorially symmetric character of MRG waves, also allows for the mixing of MRG waves with the other equatorial waves of similar characteristics (e.g., $n=2$ equatorial Rossby waves). However, it is expected that the limited wavenumber and frequency filtering range (2–10 westward wavenumbers and 3–10-day periodicity) used for extracting the meridional wind anomalies would minimise the possibility of mixing MRG waves with the $n=2$ and higher even-number meridional modes of ER waves that exhibit a low-frequency character. It has been recognised that the presence of background circulation can modulate the restoring force of equatorial waves, which in turn can cause variability in the trapping scale of equatorial waves (Zhang & Webster, 1989). Since easterly and westerly winds are observed in the tropical upper troposphere, we do expect notable differences in the trapping scales of MRG wave events. Unlike the wave projection method, the wave-fitting method treats the trapping scale as a variable parameter and it enables estimating the instantaneous value of the trapping scale (Au-Yeung & Tam, 2018). There is a possibility that the intrusion of extratropical disturbances over the westerly ducts might lead to an overestimation of the trapping scale and in turn impact the identification of MRG wave events. However, the close similarity between the MRG wave events identified using wave-fitting and wave projection methods (figure not shown), gives more confidence in the wave-fitting-based approach used in the study. A recent study by Knippertz and co-authors (2022) provides a comprehensive review on the subject of equatorial wave identification techniques.

We have identified 2390 MRG wave events at 200 hPa for the 1979–2019 period. Important dynamical properties of the MRG waves during different time instances of an MRG wave event are estimated as follows. The

maximum of meridional wind anomaly at the equator is taken as the amplitude of the MRG wave. The trapping scale is estimated from the y_0 values which show minimum error. The instantaneous wavenumber of an MRG wave event is estimated by applying Fourier transform in space. Instantaneous phase speed of an MRG wave event is estimated using the fourth-order Runge–Kutta finite difference method. Statistical significance of the differences in wave properties of the MRG wave events are tested by conducting Student's t and F tests.

The presence of extratropical disturbances over the subtropical and tropical region during MRG wave events is identified by following the methodology of Waugh and Polvani (2000). Waugh and Polvani (2000) traced the intrusion of extratropical air into the tropical region by exploiting the PV conservation in the absence of diabatic processes. By following a similar strategy, PV on the 350-K isentropic surface is used as a tracer to identify the extratropical wave intrusions during MRG wave events. Since the 350-K isentropic surface lies in the lower stratosphere in the extratropics and in the upper troposphere in the tropics (close to 200 hPa), the observation of higher PV on the 350-K isentropic surface over the tropical region suggests the presence of extratropical air in the tropics. An MRG wave event is grouped as the one associated with extratropical wave intrusions only when the criteria involving amplitude and duration of the PV at 350 K over a region where the MRG waves are active, are satisfied. It is noted that the extratropical disturbances can intrude into the tropics by taking NH and SH pathways. First, the amplitude of the PV at 350 K must be greater than 1 Potential Vorticity Unit (PVU; $1 \text{ PVU} = 10^{-6} \text{ m}^2 \cdot \text{s}^{-1} \cdot \text{kg}^{-1}$) at 15° N or 15° S within 40° east or west of the longitude at which the meridional wind of MRG waves exhibits a maximum. Second, the above condition must be satisfied for at least two days before the peak MRG wave activity and the high PV amplitude should be persistent over a longitudinal extent of 10° . If an MRG event satisfies these two criteria, such an MRG event is classified as an intrusion MRG event.

3 | RESULTS AND DISCUSSIONS

The equatorial wave characteristics, including that of MRG waves, are known to be sensitive to the mean zonal winds (Wang & Xie, 1996; Yang & Hoskins, 2016; Zhang & Webster, 1989, 1992). In addition to Doppler shifting of the intrinsic frequency, mean zonal winds can modulate the dynamics of MRG waves and restrict MRG wave activity either to the upper or the lower troposphere (Wang & Xie, 1996; Zhou & Wang, 2007). Figure 2 shows the longitude–pressure distribution of

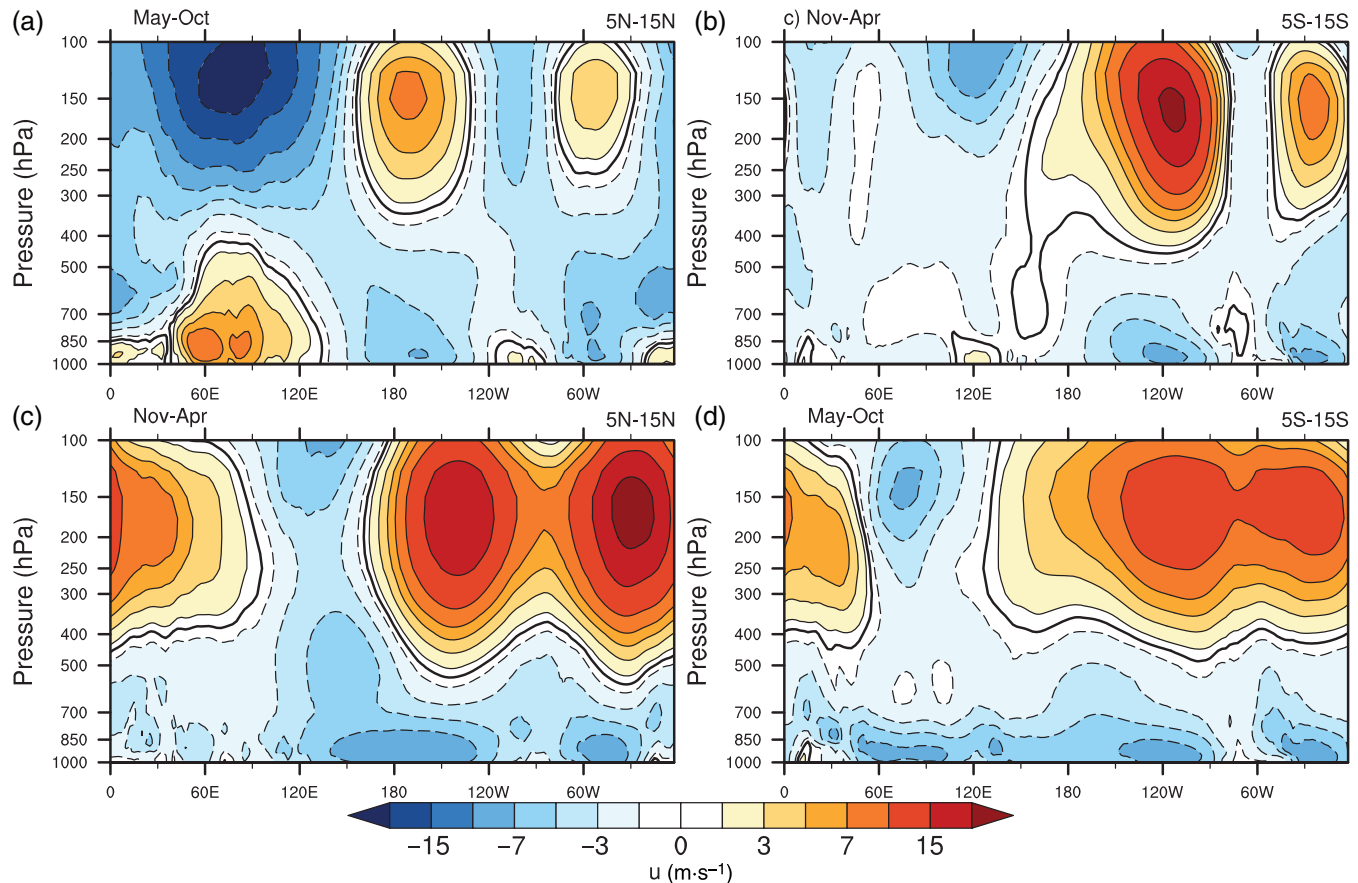


FIGURE 2 Longitude–pressure distribution of zonal winds ($\text{m}\cdot\text{s}^{-1}$) averaged over: (a) 5°N – 15°N latitudes for northern hemispheric summer seasons (May–October); (b) 5°N – 15°N latitudes for northern hemispheric winter seasons (November–April); (c) 5°S – 15°S latitudes for northern hemispheric winter seasons (November–April); and (d) 5°S – 15°S latitudes for northern hemispheric summer seasons (May–October). [Colour figure can be viewed at wileyonlinelibrary.com]

mean zonal winds averaged over the NH (5°N – 15°N) and SH (5°S – 15°S) tropical regions, during NH summer and winter seasons. Westerly ducts are known to support extratropical–tropical interactions which might result in the excitation of MRG waves (Magaña & Yanai, 1995; Randel, 1992; Webster & Holton, 1982). Several previous studies have brought out observational evidence for the NH and SH pathways of extratropical intrusions and the subsequent interactions with MRG waves (Kiladis et al., 2016; Magaña & Yanai, 1995; Yang et al., 2007; Yang & Hoskins, 2016). Well-developed westerly wind ducts are observed in the upper troposphere over the central–east Pacific and Atlantic domains in the winter hemispheres (Figure 2b,d), while the westerly wind ducts are weak and mostly confined to a smaller region in the summer hemisphere (Figure 2a,c). It implies the higher likelihood of upper-tropospheric intrusions of extratropical disturbances during winter seasons. On the other hand, in the lower troposphere, westerly winds are observed only during NH summer, its amplitude is weak and is limited to the NH IO domain. The dominance of easterly zonal wind

regime in the lower troposphere indicates that the lower troposphere is less conducive for the extratropical intrusions and the upper-tropospheric wind fields are chosen for quantifying the MRG–extratropical interactions.

Since MRG waves belong to the class of synoptic-scale variability, the synoptic-scale (3–10 day) spectral variance of zonal winds is examined for the NH (5°N – 15°N) and SH (5°S – 15°S) tropical regions during summer and winter seasons. Synoptic-scale spectral variance is estimated by computing the power spectra for multiple 90-day segments and averaging it for 3–10-day periodicity. The longitude–pressure distribution of synoptic-scale variance of zonal winds exhibits similar seasonal and geographical preferences as the zonal mean winds (Figure 3). Strong synoptic-scale variance is observed in the mid to upper troposphere mostly in the western hemisphere, and the synoptic-scale disturbances are more prominent over the winter hemisphere (Figure 3b,c). The longitude–pressure distribution of synoptic-scale variance of meridional winds is similar to that of the zonal winds (figure not shown). The regions of strong synoptic-scale wind variance represent

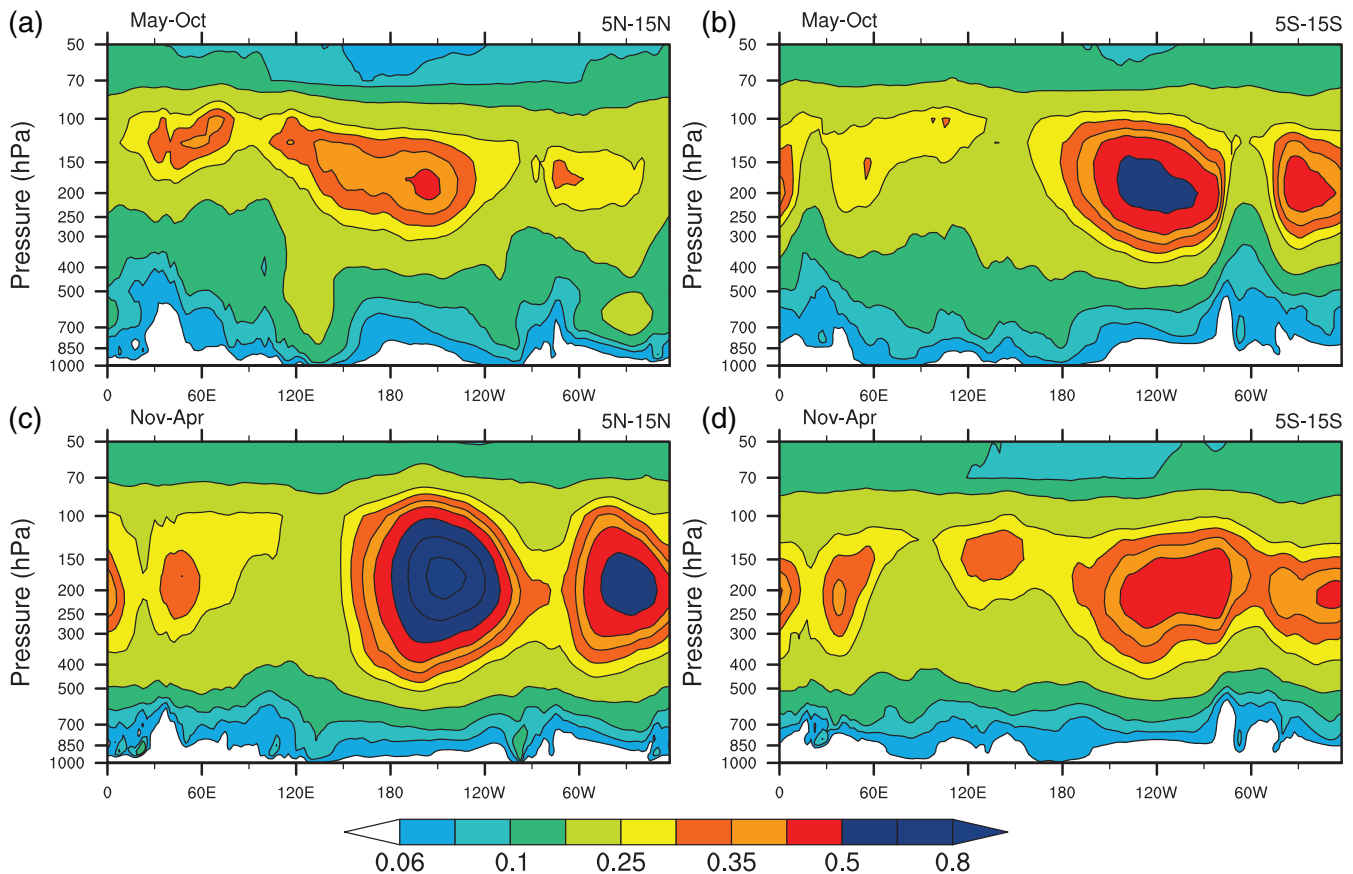


FIGURE 3 Longitude–pressure distribution of 3–10-day power spectra in zonal winds ($\text{m}^2\cdot\text{s}^{-2}$) averaged over: (a) 5°N – 15°N latitudes for northern hemispheric summer seasons (May–October); (b) 5°N – 15°N latitudes for northern hemispheric winter seasons (November–April); (c) 5°S – 15°S latitudes for southern hemispheric winter seasons (May–October); and (d) 5°S – 15°S latitudes for southern hemispheric summer seasons (November–April). [Colour figure can be viewed at wileyonlinelibrary.com]

the regions of maximum perturbation kinetic energy (PKE). The observed collocation of westerly wind ducts and the PKE maxima corroborates the findings of many previous studies (Arkin & Webster, 1985; Chen et al., 2016; Webster & Chang, 1988; Webster & Holton, 1982). The available potential energy (APE) to PKE conversions is examined by calculating the cospectra between pressure vertical velocity and temperature for NH summer and winter seasons (Figure 4). A positive co-spectral power is an indicator of APE to PKE conversions and it can be used as a measure of convection-wave coupling strength (Chen et al., 2016; Yanai et al., 2000). From the longitude–pressure distribution of synoptic-scale (3–10 day) co-spectral power (Figure 4a), it is evident that the generation of synoptic-scale motions through convection is predominant in the entire tropical upper troposphere during the summer season, while the upper-tropospheric APE to PKE conversion is weak in the western hemisphere during the winter season and it is noticeably absent over the central and east Pacific (EP) and Atlantic Ocean (AO) (Figure 4b). While the westerly wind ducts are collocated

with the synoptic-scale PKE maxima, the conspicuous absence of APE to PKE conversions over the westerly wind duct locations implies that the synoptic-scale motions over such regions are likely to be caused by dynamical processes or extratropical forcing (Charney, 1963; Holton, 2004). Recent studies based on the one-dimensional wave propagation theory rule out the variation of zonal flow and wave energy accumulation as the source of synoptic-scale PKE maxima in the western hemisphere (Hoskins & Yang, 2016; Yang & Hoskins, 2016).

In a nutshell, analysis of the mean zonal winds, synoptic-scale variance and conversion of APE to PKE implies that the excitation mechanisms of upper-tropospheric MRG waves can differ regionally and can have seasonal preferences. Examining the number of occurrences of MRG wave events as a function of longitude and months would help in understanding whether the MRG wave events exhibit any spatial and temporal clustering. Using the methodology of Au-Yeung and Tam (2018), the spatial–temporal distribution of MRG events is analysed by calculating the monthly frequency

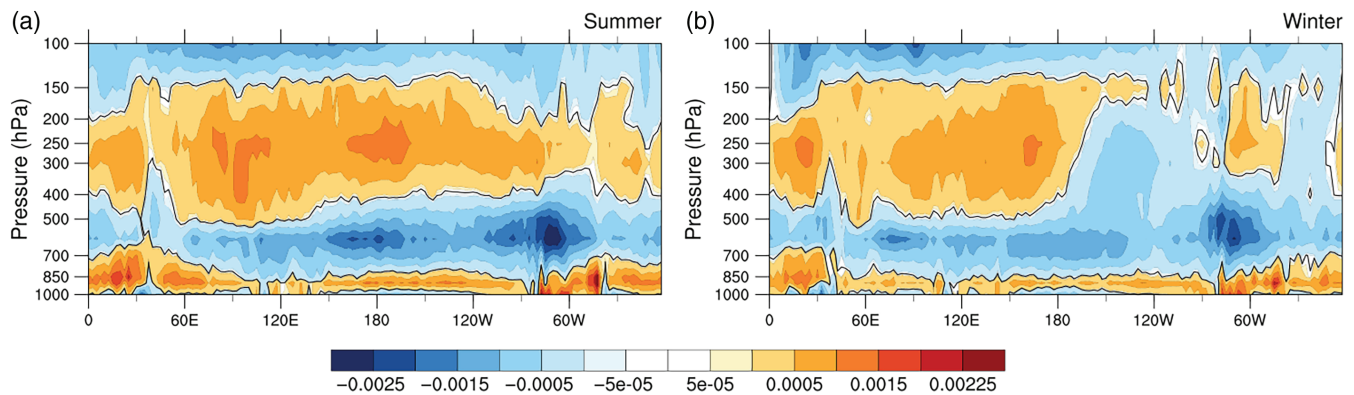


FIGURE 4 Longitude–pressure distribution of 3–10-day cospectra of temperature and vertical pressure velocity averaged over the tropics (15° S–15° N) for (a) northern hemispheric summer seasons (May–October); and (b) northern hemispheric winter seasons (November–April). Temperature and vertical pressure velocity anomalies are normalised with respect to their respective standard deviation before estimating the cross-spectra. [Colour figure can be viewed at wileyonlinelibrary.com]

of occurrence of MRG wave events in 30° longitude bins. Figure 5a shows the climatological distribution of 200-hPa MRG wave events during the months of January to December for the period 1979 to 2019. The analysis shows that MRG wave events occur over all the major ocean basins and exhibit a clear seasonality. More than 60% of the total MRG wave events are observed during NH summer season. A major fraction of MRG wave events over the Indian, Pacific and Atlantic Ocean domains occur during NH summer and secondary maxima of MRG wave events are observed during the NH winter season over the central–east Pacific and IO domains.

Based on the spatial distribution of MRG wave events, we have identified four preferred domains of MRG activity – IO (covers 50° E to 109.5° E), WP (covers 110° E to 179.5° E), EP (covers 180° E to 89.5° W) and AO (covers 90° W to 49.5° E) domains. The statistics of MRG wave events over different domains are summarised in Table 1. As indicated by many previous studies, about 65% of the MRG wave events are observed over the EP and AO domains (Kiladis et al., 2016; Randel, 1992; Suhas et al., 2020; Yang & Hoskins, 2016). It is interesting to note that about two-thirds of the total MRG wave events are equally distributed over the EP and AO domains. The robustness of the method of identification of MRG wave events is confirmed by examining the MRG wave event composites of dynamical fields over the four domains. Figure 6 shows the MRG wave event composites of 3–15-day filtered 200-hPa wind and stream function anomalies over the IO, WP, EP and AO domains. The composites are constructed with respect to the reference longitudes around which the largest number of MRG wave events is observed in each domain. The selected reference longitude for the IO, WP, EP and AO domains are 70° E, 120° E, 135° W and 40° W, respectively. Although the horizontal structure of the MRG waves appears to be similar

over all the domains, some differences in their properties are observed over the different domains. For example, the amplitude, wavelength and meridional extent of the MRG waves are larger over the EP domain (Figure 6c). It is also evident that the convection–circulation coupling strength of MRG wave events varies over different domains. The observation of stronger convection–circulation coupling over the IO and WP domains suggests the prevalence of moist dynamics.

The differences in important wave properties such as the amplitude, wavenumber, trapping scale, phase speed and duration of the individual MRG wave events over all the four domains are further evaluated. Figure 7 shows the range, mean and 90th- and 10th-percentile values of the MRG wave properties over the different domains during summer and winter seasons. Clear seasonality in the major MRG wave properties is observed across all domains. The amplitude of MRG waves is higher during NH summer and the largest- and smallest-amplitude MRG wave events are observed over the EP and AO domain, respectively (Figure 7a). It is noted that the difference in the distribution of amplitude of the MRG waves during summer and winter over the WP and AO domains are statistically significant at the 95% confidence level. In other words, compared to the other domains, the amplitudes of the MRG wave events over the IO and EP domain exhibit less seasonal variability. Summer–winter differences in the distribution of wavenumbers over the EP and AO domains are statistically significant at the 90% confidence level. The MRG wave events exhibit a shorter wavelength (larger wavenumber) during the NH summer season (Figure 7b). The MRG convection–circulation coupling strength is stronger during the NH summer (Suhas et al., 2020) and studies (Aiyer & Molinari, 2003; Zhou & Wang, 2007) indicate that the MRG scale selection is strongly tied to the active feedback between frictional

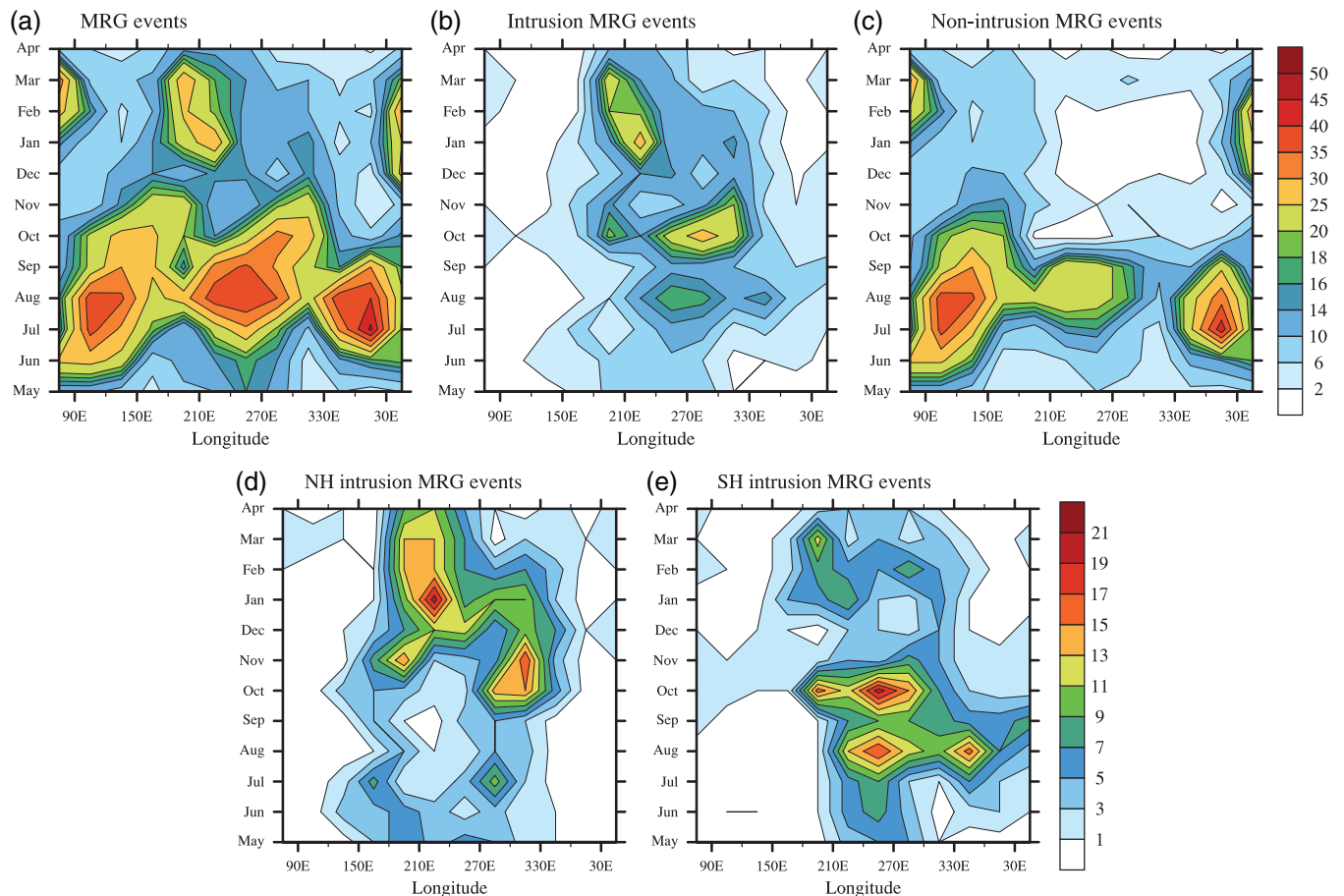


FIGURE 5 Month-longitude distribution of number of occurrences of mixed Rossby-Gravity (MRG) wave events at 200 hPa observed for the period 1979–2019. (a) Total number of occurrences of MRG wave events; (b) number of occurrences of MRG wave events that are associated with the intrusions of extratropical disturbances; (c) number of occurrences of MRG wave events that are not associated with the intrusions of extratropical disturbances, number of occurrences of MRG wave events that are associated with the intrusions of extratropical wave disturbances via (d) the northern hemisphere and (e) the southern hemisphere. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/qj.4668)]

TABLE 1 Number of occurrence of intrusion mixed Rossby-Gravity (MRG) events, non-intrusion mixed Rossby-Gravity (MRG) events and total MRG events observed over the Indian Ocean, West Pacific Ocean, central-east Pacific Ocean and Atlantic Ocean domains for the period 1979–2019.

MRG event category	Indian Ocean (50° E–110° E)	West Pacific Ocean (110° E–180° E)	central-east Pacific Ocean (180° E–90° W)	Atlantic Ocean (90° W–50° E)	Total MRG events
Intrusion MRG events	29	77	480	314	900 (37%)
Non-intrusion MRG events	361	387	276	466	1490 (63%)
Total MRG events	390 (16%)	464 (19%)	756 (32%)	780 (33%)	2390

Note: Fraction (%) of the total MRG events observed over different domains are shown in parentheses.

convergence and convective processes. The shorter wavelength observed during NH summer might be related to a stronger feedback between frictional convergence and convection (Zhou & Wang, 2007). The relatively smaller trapping scale of MRG wave events over the IO and

WP domains (Figure 7c) suggests that probably convection plays a dominant role in triggering the MRG waves over these domains (Figure 6a,b). Tropical organised convection makes the atmosphere less stable and decreases the effective equivalent depth which in turn reduces the

FIGURE 6

Composites of mixed Rossby–Gravity (MRG) wave events constructed using 3–15-day bandpass-filtered 200 hPa wind ($\text{m}\cdot\text{s}^{-1}$) and outgoing longwave radiation (OLR; $\text{W}\cdot\text{m}^{-2}$) anomalies over: (a) the Indian Ocean (IO); (b) West Pacific Ocean (WP); (c) central–east Pacific Ocean (EP); and (d) Atlantic Ocean (AO) domains. While compositing the wind and OLR anomalies are longitudinally translated to a reference longitude corresponding to maximum occurrence of MRG wave events for each domain. [Colour figure can be viewed at wileyonlinelibrary.com]

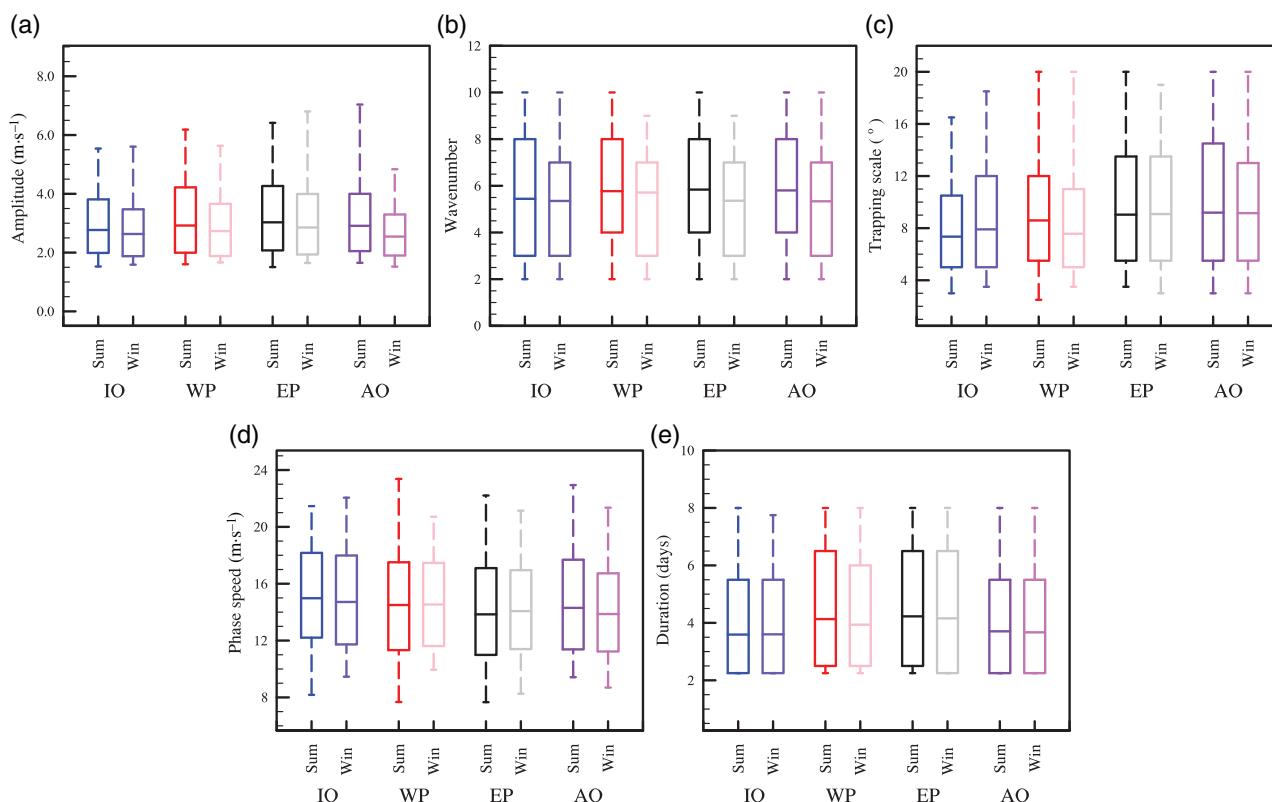
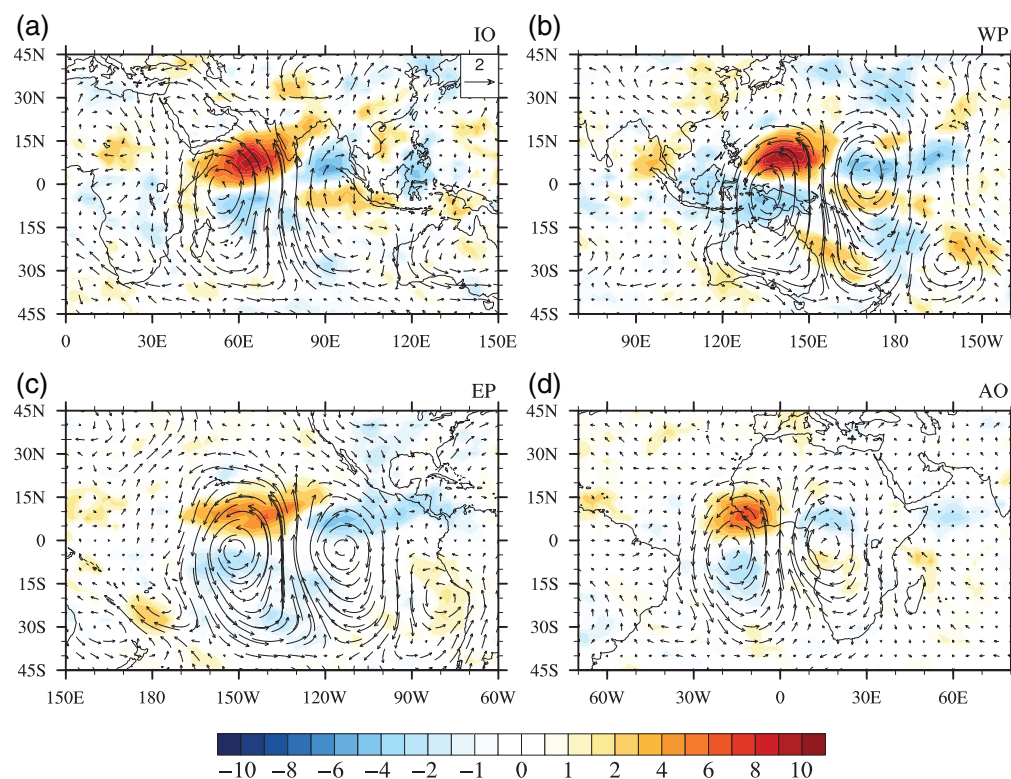


FIGURE 7 Range (maxima, minima), 90th percentile, mean, and 10th percentile of: (a) amplitude ($\text{m}\cdot\text{s}^{-1}$); (b) wavenumber; (c) trapping scale ($^{\circ}$); (d) phase speed ($\text{m}\cdot\text{s}^{-1}$); and (e) duration (days) of mixed Rossby–Gravity (MRG) wave events observed over the Indian Ocean (IO, blue), West Pacific (WP, red), central–east Pacific (EP, black) and Atlantic Ocean (AO, purple) domains during northern hemispheric summer (Sum, dark colour) and northern hemispheric winter (Win, light colour).

trapping scale of wave disturbances (Kiladis et al., 2009). A relatively larger trapping scale is observed over the less convectively active EP and AO domains (Figure 7c). Zonal mean winds and extratropical forcing also play a role in determining the trapping scale of MRG waves (Zhang & Webster, 1989, 1992). The larger trapping scale might be indicative of the role of wave–mean flow interactions and intrusions of extratropical disturbances in triggering MRG wave events over these domains. The MRG wave phase speeds range from 12 to 17 m·s^{−1} and in contrast to the other parameters, they do not show much seasonality and variations across the different domains (Figure 7d). The duration of MRG wave events is also not very sensitive to the season. However, longer-duration MRG wave events are observed over the Pacific domain (Figure 7e).

The possible link between MRG wave events and intrusions of extratropical disturbances are investigated by examining the PV on the 350-K isentropic surface. Using the intrusion identification methodology described above, we find that about 37% of the MRG wave events are associated with intrusions of extratropical disturbances (Table 1) and the majority (54%) of such MRG wave events occur during NH winter (Table 2). The longitudinal distribution of monthly frequency of occurrence of MRG wave events associated with and without intrusions of extratropical disturbances is shown in Figure 5b,c. It is indicated in Table 1 that a significant fraction of intrusion MRG wave events (88%) occurs over the EP and AO domains, and about half of the MRG wave events over these domains are associated with intrusions of extratropical disturbances. On the other hand, a large fraction of the MRG wave events (86%) over the IO and WP domains is not associated with intrusions of extratropical disturbances, consistent with the existence of an upper-level easterly basic state that would preclude the propagation of Rossby wave energy into those regions.

Intrusion MRG wave events are further classified as NH (SH) intrusion MRG wave events in the case when extratropical air intrusions occur in the NH (SH), respectively. In rare cases, when intrusions are observed in both hemispheres, the event is classified based on the strength of the PV. The statistics and seasonality of the NH and SH intrusion MRG events and non-intrusion MRG events over the four domains are summarised in Table 2. The longitudinal distribution of monthly frequency of occurrence of NH and SH intrusion MRG wave events is shown in Figure 5d,e. The analysis reveals that intrusion of extratropical disturbances from both the NH and the SH is equally important in the genesis of MRG wave events. Except Yang and Hoskins (2016) and Kiladis et al. (2016), all other past studies only reported the association between the intrusion of extratropical disturbances via SH and MRG wave events (e.g., Magaña & Yanai, 1995; Suhas et al., 2020). The NH intrusion MRG wave events exhibit a

TABLE 2 Number of occurrences of northern hemispheric (NH) intrusion mixed Rossby–Gravity (MRG) events, SH intrusion events, co-occurrence of NH and SH intrusion MRG events and non-intrusion MRG events observed during NH summer and winter seasons for the period 1979–2019.

MRG event category	Indian Ocean (50° E–110° E)				West Pacific Ocean (110° E–180° E)				central-east Pacific Ocean (180° E–90° W)				Atlantic Ocean (90° W–50° E)			
	NH		SH		NH		SH		NH		SH		NH		SH	
	intrusion	non-intrusion	intrusion	non-intrusion	intrusion	non-intrusion	intrusion	non-intrusion	intrusion	non-intrusion	intrusion	non-intrusion	intrusion	non-intrusion	intrusion	non-intrusion
Intrusion: NH summer	0	0	8	10	36	0	3	12	68	154	0	6	50	113	10	448 (19%)
Intrusion: NH winter	10	1	22	266	22	4	12	162	73	17	85	44	12	452 (19%)		
Non-intrusion: NH summer	209								220		333					1028 (43%)
Non-intrusion: NH winter	152		121		464 (19%)				56		133					462 (19%)
Total	390 (16%)								756 (32%)		780 (33%)					2390

maxima over the EP during December–January and over the AO during October–November. Likewise, a large number of SH intrusion MRG wave events is observed during SH winter. Consistent with the inferences made from the distribution of mean zonal winds, the winter hemisphere with the westerly wind ducts seem to provide a favourable environment for the occurrence of intrusion MRG wave events (Figure 2b,d). About 70% of the intrusion MRG wave events over the EP and AO domains are observed in the winter hemisphere.

Spatial–temporal evolution of the upper-tropospheric tropical and extratropical dynamical fields during MRG wave events is examined for further validating the intrusion and non-intrusion MRG wave event classification. Lead–lag composites of dynamical fields associated with MRG wave events are constructed by identifying the day and longitudinal location of the amplitude maxima corresponding to each MRG wave event and translating the longitudinal locations to a reference longitudinal location and separately averaging the dynamical fields two days before, two days after and on the day of the MRG amplitude maxima. Figure 8 shows the lead–lag composites of 3–15-day filtered wind and stream function anomalies at 200 hPa associated with the intrusion MRG wave events over the EP domain. Statistical significance of the composites

of MRG wave events are tested by conducting *t*-tests. Northwest–southeast-oriented stream function anomalies propagating from the NH extratropical to tropical region are found in the composites of NH intrusion MRG wave events during summer and winter (Figure 8a–f). At the same time, southwest–northeast-oriented eastward-propagating wave trains composed of alternate cyclonic and anticyclonic wind anomalies arching towards the equator are found in the SH in the composites of SH intrusion MRG wave events during summer and winter (Figure 8g–l). The observation of differences in the amplitude of the intruding extratropical disturbances two days before (stronger) and two days after (weaker) the MRG wave event amplitude maxima indicates the role of extratropical disturbances in triggering the MRG wave events. Weakening of the intruding extratropical disturbances a few days after the MRG amplitude maxima could be explained based on the PV–convection relationship. Past studies showed the occurrence of convection downstream of the intruding extratropical disturbances over the EP (Kiladis, 1998). Diabatic heating associated with convection can induce anticyclonic anomalies in the upper level which in turn can weaken the pre-existing cyclonic anomalies through destructive interference and thereby weaken the wave trains.

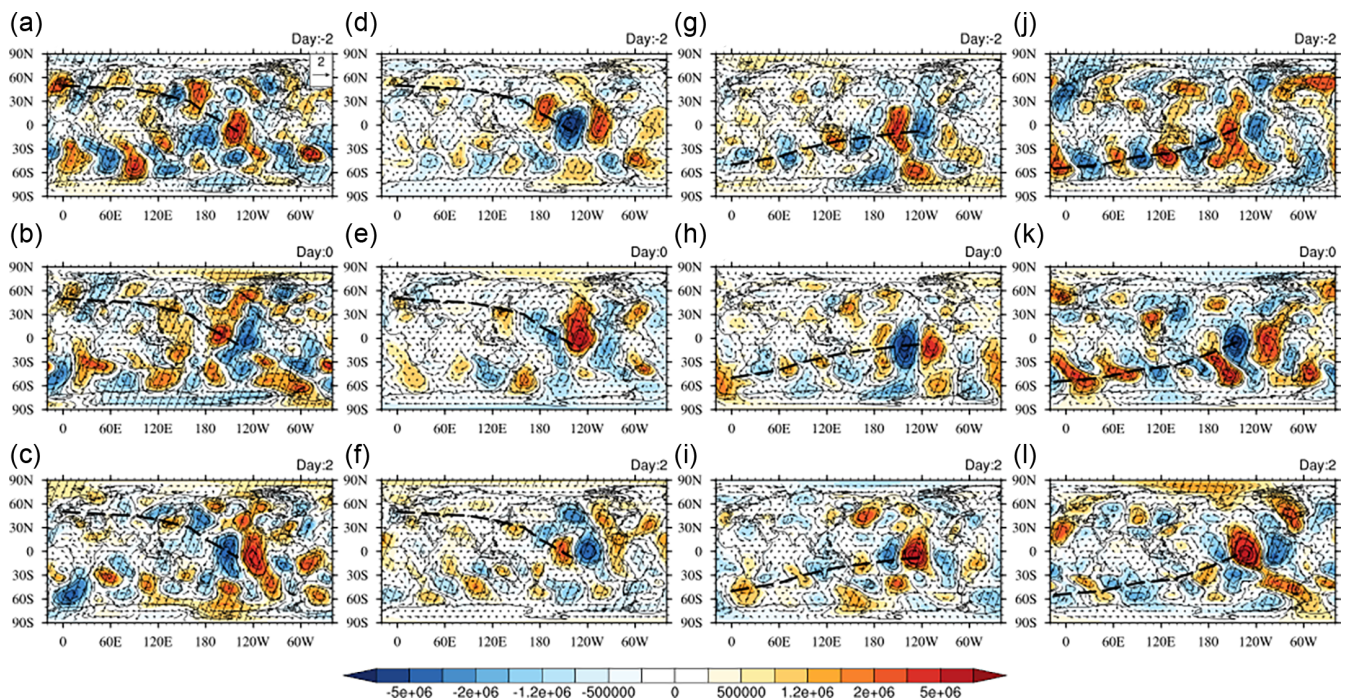


FIGURE 8 Lead–lag composites of intrusion MRG wave events over the central-east Pacific Ocean domain, constructed using 3–15-day bandpass-filtered 200 hPa wind ($\text{m}\cdot\text{s}^{-1}$) and stream function ($\text{m}^2\cdot\text{s}^{-1}$) anomalies. Positive (negative) lag represents the number of days after (before) the maximum mixed Rossby–Gravity (MRG) wave amplitude. (a–c) NH intrusion MRG wave events during NH summer; (d–f) NH intrusion MRG wave events during NH winter; (g–i) SH intrusion MRG wave events during NH summer; and (j–l) SH intrusion MRG wave events during NH winter. Black dashed curves highlight the intrusions of extratropical disturbances. Stippled regions are statistically significant at the 95% confidence level. [Colour figure can be viewed at wileyonlinelibrary.com]

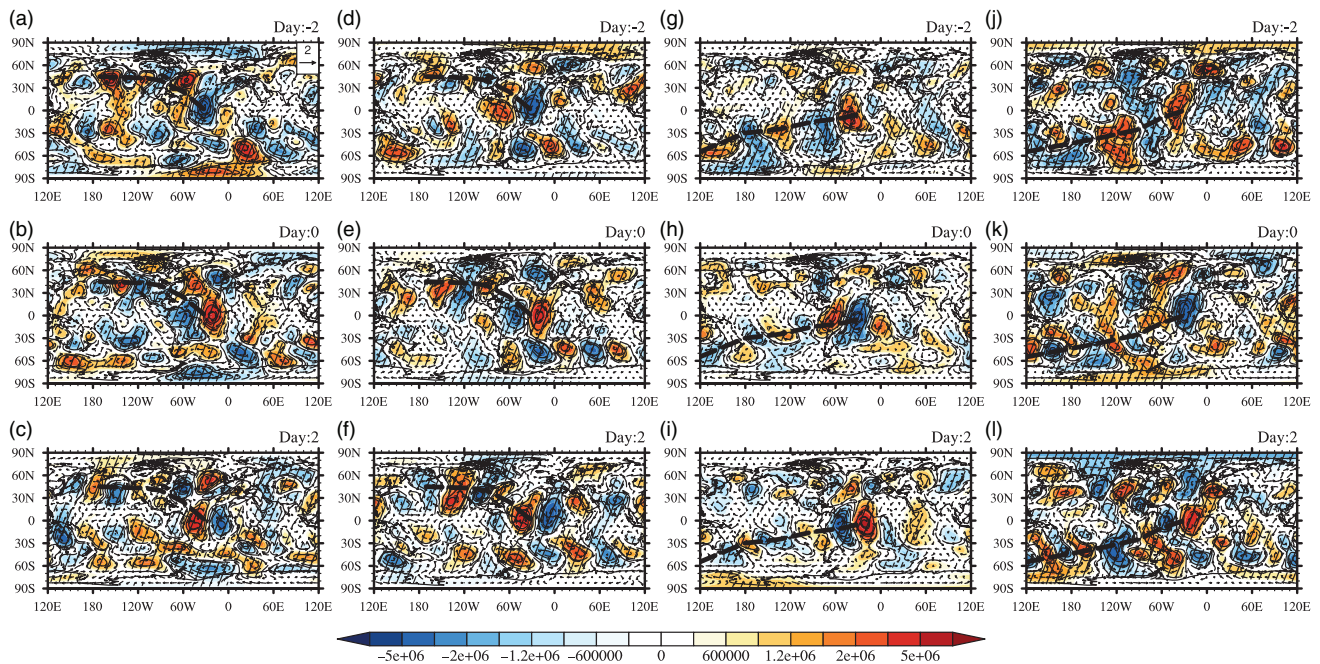


FIGURE 9 Lead-lag composites of intrusion mixed Rossby-Gravity (MRG) wave events over the Atlantic Ocean domain, constructed using 3–15-day bandpass-filtered 200 hPa wind ($\text{m}\cdot\text{s}^{-1}$) and stream function ($\text{m}^2\cdot\text{s}^{-1}$) anomalies. Positive (negative) lag represents the number of days after (before) the maximum MRG wave amplitude. (a–c) NH intrusion MRG wave events during NH summer. (d–f) NH intrusion MRG wave events during NH winter; (g–i) SH intrusion MRG wave events during NH summer; and (j–l) SH intrusion MRG wave events during NH winter. Black dashed curves highlight the intrusions of extratropical disturbances. Stippled regions are statistically significant at the 95% confidence level. [Colour figure can be viewed at wileyonlinelibrary.com]

Similar to life cycle composites of the intrusion MRG wave events over the EP domain, northwest-southeast-oriented and southwest-northeast-oriented equatorward and eastward-propagating extratropical disturbances are observed over the AO domain during summer and winter (Figure 9). It seems that the inter-hemispheric propagation of extratropical disturbances is more likely over the AO domain (Figure 9a–f). It can be deduced from the lead-lag composites that the intruding extratropical disturbances exhibit wavenumbers in the range of 4–7 (Figures 8, 9).

Figure 10 represents the composites of 3–15-day filtered 200-hPa wind and stream function anomalies of non-intrusion MRG wave events over the IO, WP, EP and AO domains during summer and winter. The composites are created for the day of the maximum MRG amplitude of non-intrusion MRG wave events. As expected, the composites of MRG non-intrusion wave events show that extratropical disturbances are either absent or weak and do not intrude into the tropics during NH summer and winter over all the domains (Figure 10). The results of the composite analyses for intrusion and non-intrusion MRG wave events validate the robustness of the method employed in classifying the MRG wave events.

Some previous studies based on idealised modelling experiments argued that lateral forcing and different

configurations of mean winds can induce differences in the properties of MRG waves (Mak, 1969; Wang & Xie, 1996; Zhang & Webster, 1992). To explore whether the intrusion and non-intrusion MRG wave events exhibit notable differences in their properties, the statistical modes of the amplitude, wavenumber, trapping scale, phase speed and duration of the two groups of MRG wave events are examined during summer and winter (Figure 11). The analysis is restricted to the EP and AO domains since approx. 88% of the intrusion MRG wave events are observed over these domains. The non-intrusion MRG wave events exhibit a larger amplitude during the summer season compared to winter over both domains (Figure 11a,f), consistent with the larger MRG amplitude observed during summer (Figure 7a). Over the EP domain, where the largest-amplitude MRG wave events are observed, during the winter season, intrusion MRG wave events exhibit a larger amplitude than the non-intrusion MRG wave events and during the summer season, the amplitude of non-intrusion MRG wave events exceeds that of intrusion MRG wave events. This explains the low seasonal variability in MRG amplitude over the EP domain. Zhang and Webster (1992) hypothesised that the MRG waves would exhibit stronger amplitudes in the regions of westerly mean zonal winds, under the influence of lateral intrusions. As seen in Figure 2,

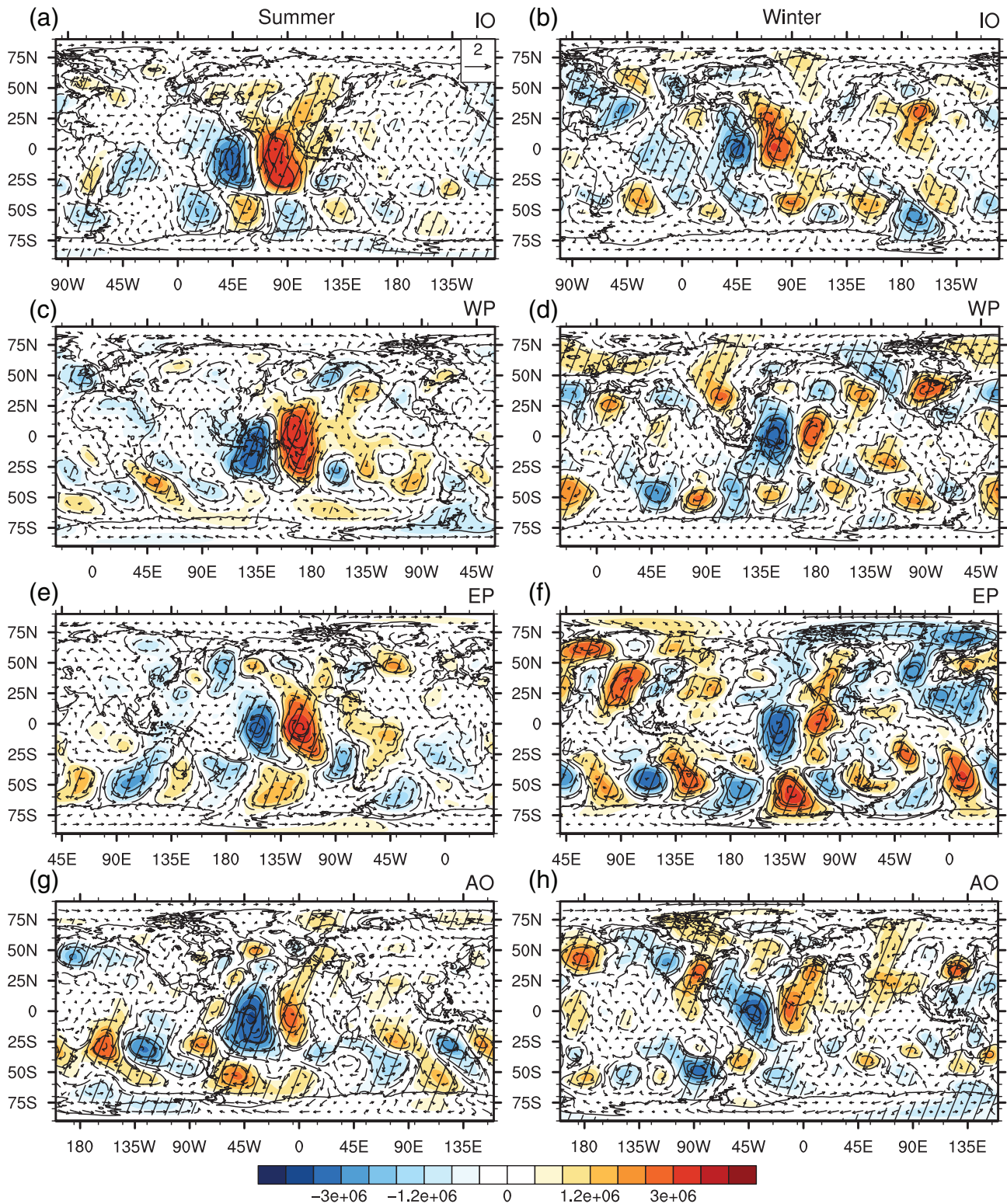


FIGURE 10 Composite non-intrusion mixed Rossby-Gravity (MRG) wave events constructed using 3–15-day bandpass-filtered 200 hPa wind ($\text{m}\cdot\text{s}^{-1}$) and stream function ($\text{m}^2\cdot\text{s}^{-1}$) anomalies over the composites of 3–15-day bandpass-filtered 200 hPa wind ($\text{m}\cdot\text{s}^{-1}$) and stream function ($\text{m}^2\cdot\text{s}^{-1}$) anomalies of non-intrusion MRG events occurring over: (a,b) the Indian Ocean; (c,d) West Pacific; (e,f) central-east Pacific; and (g,h) Atlantic Ocean domains during NH summer (left column) and winter (right column). Stippled regions are statistically significant at the 95% confidence level. [Colour figure can be viewed at wileyonlinelibrary.com]

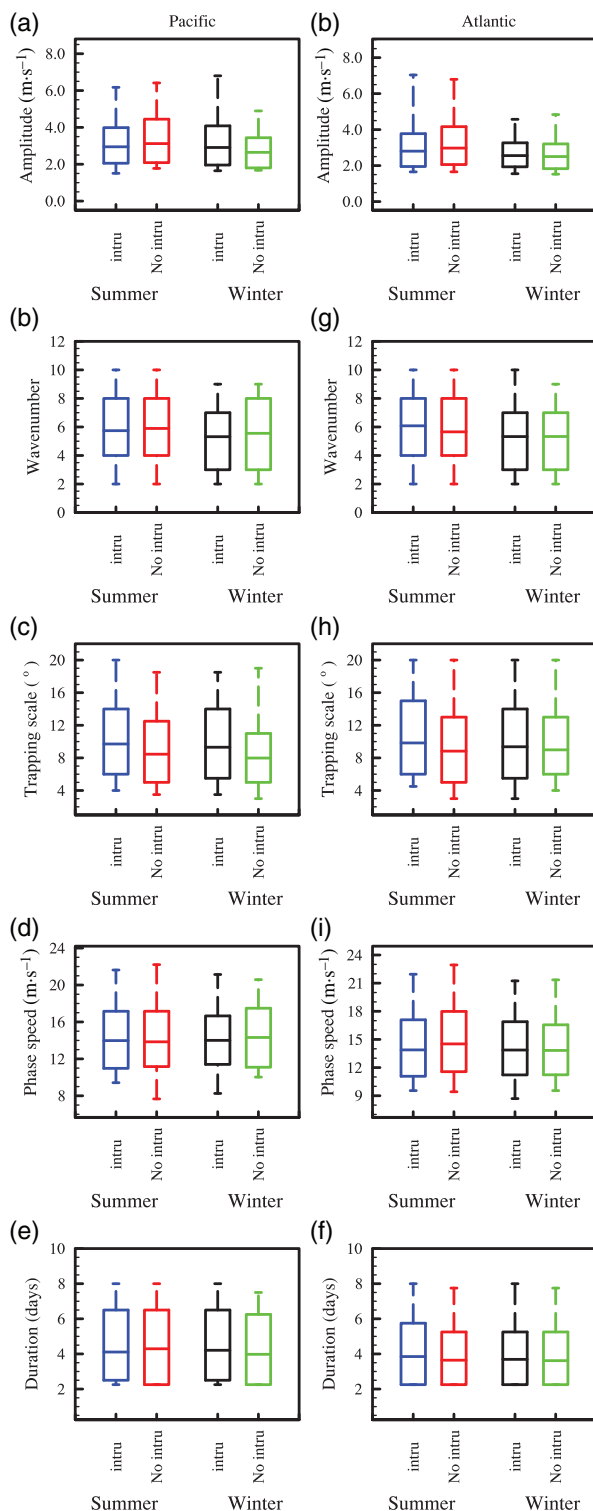


FIGURE 11 Range (maxima, minima), 90th percentile, mean, and 10th percentile of: (a,f) amplitude ($\text{m}\cdot\text{s}^{-1}$); (b,g) wavenumber; (c,h) trapping scale ($^{\circ}$); (d,i) phase speed ($\text{m}\cdot\text{s}^{-1}$); and (e,j) duration (days) of intrusion and non-intrusion mixed Rossby–Gravity (MRG) wave events occurring over the central–east Pacific (left column) and Atlantic Ocean (right column) domains during NH summer and winter. Grey and black boxes represent intrusion and non-intrusion MRG wave events, respectively for NH summers and NH winters. [Colour figure can be viewed at wileyonlinelibrary.com]

the upper-tropospheric westerly wind ducts are well developed over the EP and AO domains only during the winter season. Consistent with this, the amplitude of intrusion MRG wave events is higher than that of the non-intrusion MRG wave events over the EP domain during the winter season (statistically significant at the 95% confidence level) (Figure 11a). Over the AO domain, where the smallest-amplitude MRG wave events are observed, the summer intrusion MRG wave events exhibit a larger amplitude compared to their winter counterparts (Figure 11f). Even though the MRG wave events prefer to have higher wavenumbers during NH summer over the EP and AO domains, the wavenumbers of intrusion and non-intrusion MRG wave events do not exhibit a clear separation except over the EP domain during NH winter (Figure 11b,g). Unlike the distribution of amplitudes and wavenumbers, statistical modes of trapping scales of intrusion and non-intrusion MRG wave events exhibit a clear separation over the EP and AO domains and it is evident from Figure 11c,h that the intrusion MRG events are less trapped to the equator (statistically significant at the 90% confidence level). However, the phase speeds and duration of the intrusion and non-intrusion MRG wave events do not show a noticeable difference in their distributions over the EP and AO domains (Figure 11d,e,i,j).

The whisker plots provide some insights on the role of extratropical disturbances in the genesis of MRG waves over the EP. According to linear wave theory, lateral forcing by eastward-moving extratropical disturbances cannot cause resonance in westward-propagating MRG waves (Yang & Hoskins, 2016). However, contrasting differences in the wave properties of intrusion and non-intrusion MRG wave events over the EP during NH winter indicate the possibility of resonant interactions. For example, the intrusion MRG wave events have a stronger amplitude and a larger meridional extent compared to non-intrusion MRG wave events (statistically significant at the 90% confidence level) (Figure 11a,c). The intrusion MRG wave events also exhibit a spatial scale similar to that of the extratropical disturbances (wavenumber 4–7) implying that resonant interactions may be a primary mechanism for the genesis of MRG wave events during NH winter. During NH summer, on the other hand, extratropical disturbances seemingly act as a catalytic factor in intensifying the MRG wave events. Past studies have shown observational evidence for the decisive role of intrusion of extratropical disturbances in determining the strength of the MRG waves during NH summer (Magaña & Yanai, 1995; Suhas et al., 2020). It can be deduced from the shorter wavelengths, smaller trapping scales (Figure 11b,c), and stronger convection–circulation coupling strength (Suhas et al., 2020) exhibited by the MRG wave events during NH summer that they could be primarily triggered by convective processes. However, the

observed weakening of the strength of MRG waves in the absence of intrusion of extratropical disturbances via the SH suggests that lateral forcing may be instrumental in amplifying their amplitude.

4 | SUMMARY AND CONCLUSIONS

In this study we have conducted a survey of westward-propagating MRG wave events at 200 hPa and quantified their association with the intrusions of extratropical disturbances using ERA-I reanalysis data for the period 1979–2019. MRG wave events are identified by imposing the theoretical meridional structure of MRG waves onto the equatorial meridional wind anomalies (Au-Yeung & Tam, 2018). The method is ideal for capturing the episodic nature of MRG waves and it also provides the instantaneous values of its amplitude, phase speed, wavenumber and trapping scale. We have identified 2390 MRG wave events in the upper troposphere and the majority (61%) of them occurred during the months of May–October (Figure 5). As pointed out by many previous studies, MRG wave events over the EP and AO domains account for a major chunk (65%) of the total MRG wave events (Table 1). Statistical modes of wave properties of the MRG wave events over the IO, western Pacific, EP and AO reveal that not only the frequency of occurrence but also the amplitude, wavenumber and trapping scale of the MRG wave events exhibit a clear seasonality (Figure 7). MRG wave events exhibit higher amplitudes and shorter wavelengths during the NH summer. MRG wave events observed over the IO domain are more closely trapped to the equator during NH summer. The MRG wave events observed in the western hemisphere are lesser trapped to the equator and it is also noted that the trapping scales do not exhibit significant seasonal variability. The phase speeds of the MRG wave events do not exhibit significant variability.

MRG wave events are classified as intrusion MRG wave events if the PV on the 350-K isentropic surface at 15° N/S latitude in the spatial-temporal neighbourhood of the MRG wave events is greater than 1 PVU. Such events are further classified as NH intrusion or SH intrusion MRG wave events based on the hemisphere where the intrusions of extratropical disturbances are observed. We find that 37% of the MRG wave events are associated with intrusions of extratropical disturbances and a large majority (88%) of such events occur over the EP and AO domains. Larger numbers of NH (SH) intrusion MRG wave events are found to occur during the NH (SH) winter. Although the intrusions of extratropical disturbances occur in both hemispheres over the EP and AO domains, it is interesting to note that nearly 70% of the intrusions of extratropical disturbances associated with

the MRG wave events occur in the winter hemisphere. The predominant preference of the winter hemispheric route of intruding extratropical disturbances corroborates the extratropical–tropical interaction theory of Webster and Holton (1982). Detailed analysis reveals that the amplitude, wavenumber and trapping scale of intrusion and non-intrusion MRG wave events are distinctly different only over the EP during NH winter (Figure 11). The intrusion MRG wave events exhibit higher amplitudes, lower wavenumbers, and larger trapping scales compared to the non-intrusion MRG wave events. Coherent evolution of extratropical and tropical disturbances is evident in the intrusion MRG wave events life cycle composites (Figures 8, 9) validating the classification of MRG wave events into intrusion and non-intrusion categories. Equatorward-propagating extratropical disturbances show northwest–southeast and southwest–northeast tilt in the NH and the SH, respectively. It is inferred from the life cycle composites that extratropical disturbances associated with the MRG wave events have wavenumbers in the range of 4–7. Stronger extratropical disturbances a few days before the MRG amplitude maxima and weakening of extratropical disturbances a few days after the MRG amplitude maxima indicate the possibility that such MRG wave events might be forced by the extratropical disturbances.

The study reveals that 37% of the MRG wave events are associated with intrusions of extratropical disturbances. It implies that external forcing of MRG waves is as important as their in-situ genesis mechanisms. Even though the diagnostic analyses may be insufficient to clearly establish the impact of lateral forcing via extratropical disturbances in exciting MRG wave events, probable scenarios can be deduced from the analyses. The observation of stronger amplitudes and larger trapping scales exhibited by intrusion MRG wave events and the similarity in spatial scales of the MRG wave events and extratropical disturbances suggest the role of resonant interactions between these two modes over the EP during NH winter. Past studies showed that the intrusions of extratropical disturbances is a crucial factor governing the strength of MRG waves during NH summer and fall (Magaña & Yanai, 1995; Suhas et al., 2020). However, the absence of a clear difference in the amplitudes of intrusion and non-intrusion MRG wave events over the EP and AO domains during NH summer rules out the possibility of resonant interactions. It could be possible that the MRG waves are excited by random convective perturbations over the EP and Atlantic domains which in turn might be amplified by the intruding extratropical disturbances (Salby and Garcia, 1987). Without such amplification by intrusions of extratropical disturbances, the growth of the MRG perturbations induced by convective processes may be limited.

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DATA AVAILABILITY STATEMENT

All the data used in this study are freely available from public data repositories. ERA-interim reanalysis data (Dee et al., 2011) can be accessed from: <https://apps.ecmwf.int/datasets/data/interim-full-daily/levtype=sfc/>. Daily NOAA interpolated OLR data (Liebmann & Hendon, 1990) can be accessed from: https://psl.noaa.gov/data/gridded/data.interp_OLR.html.

ENDNOTE

¹ Although the wave-fitting methodology can be used as an approximation of wave projection methodology in identifying the MRG wave events, it should be noted that the wave-fitting methodology works only with the meridional winds since the zonal winds of MRG waves are identified by two parabolic cylindrical functions.

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